Yuning Wu

Carnegie Mellon University
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Research Interest
Multi-Agent Robotic System, Collective Robotic Construction, Robotic Learning
Machine Learning & Robotics

Education

Carnegie Mellon University Pittsburgh, PA
Advisor: Daniel C. Llach
08/19 – Ph.D. Track, Computational Design, 3.76/4.33.

Carnegie Mellon University Pittsburgh, PA
Advisor: Katia Sycara, Liuchang Liu
08/19 – M.S., Machine Learning, 3.8/4.33.

Politecnico di Torino Turin, Italy
09/15 – 07/16 (Double Degree with Tsinghua) M.S., Architecture and Construction, 28/30.

Tsinghua University Beijing, China
08/14 – 07/17 M.Arch., Architecture (Construction Robots and Digital Fabrication), top 10%.
08/11 – 07/14 B.Eng., B.Arch., Architecture, top 10%.
08/10 – 07/11 Mechanical Engineering (Energy and Power System), top 1.

Research Experience

CMU, CoDe Lab Pittsburgh, PA
Advisor: Daniel C. Llach
08/19 – Multi-drone construction, say something, Distinctive Graduation Dissertation by Tsinghua University Distinctive Graduation Dissertation by Tsinghua University Distinctive Graduation Dissertation by Tsinghua University Distinctive Graduation Dissertation by Tsinghua University

CMU, Intelligent Control Lab Pittsburgh, PA
Advisor: Liuchang Liu
06/20 – Multi-drone construction, say something, Distinctive Graduation Dissertation by Tsinghua University Distinctive Graduation Dissertation by Tsinghua University Distinctive Graduation Dissertation by Tsinghua University Distinctive Graduation Dissertation by Tsinghua

CMU, Advanced Agent – Robotics Technology Lab Pittsburgh, PA
Advisor: Katia Sycara
07/20 – Multi-drone construction, say something, Distinctive Graduation Dissertation by Tsinghua University Distinctive Graduation Dissertation by Tsinghua University Distinctive Graduation Dissertation by Tsinghua University Distinctive Graduation Dissertation by Tsinghua

Honors & Awards
Lego Scholarship (Lego Foundation, CMU. 2019)
Distinctive Graduation Dissertation (Tsinghua University. 2017),
Delegate Student Speaker at University Inauguration (Politecnico di Torino. 2016),

Champion of Student Forimula Racing Challenge (Audi & Tsinghua University. 2014)
Fellow of Tsinghua Siyuan Elite Student Program (Tsinghua University. 2010 - 2014)

Publications[1]

Zhihao Fang*, Ardavan Bidgoli*, Yuning Wu*, Ammar Hassonjee, and Daniel C. Llach. Semi-Autonomous Construction in Dynamic Environments. In *Proceedings of the 40th Annual Conference of the Association for Computer Aided Design in Architecture (ACADIA-20)*, 2020. *Pending.*

Tianbo Hao*, Yuning Wu*, and Phillip B. Gibbons. Semi-Autonomous Construction in Dynamic Environments. In *Proceedings of the 40th Annual Conference of the Association for Computer Aided Design in Architecture (ACADIA-20)*, 2020. *Pending.*

Tianbo Hao*, Yuning Wu*, Changliu Liu, and Katia Sycara. Semi-Autonomous Construction in Dynamic Environments. In *Proceedings of the 40th Annual Conference of the Association for Computer Aided Design in Architecture (ACADIA-20)*, 2020. *Pending.*

Yuning Wu. A-DNN: From Additive Model to Additive Dense Neural Network. 2019. *Unpublished*.

Presentations

O. Akin and Y. Wu. State Space Paradox in Computational Research of Form Creation. In *30th Anniversary "Heron Island" Conference on Computational and Cognitive Models (HI'19)*, 2019.

Project

Robust, scalable multi-agent drone construction with a reinforcement learning approach.
CMU CoDe Lab
Develop a robust semi-autonomous framework under which drones could collaborate in completing various construction forms and tasks (pick & place, weave, weld, etc.) in dynamic environment. Mainly use reinforcement learning In path planning and task allocation, algorithms include DQN, MADDPG, A3C, REINFORCE, etc. The general workflow is formed by physical environment modeling and simulation (Unity), RL training and inference (OpenAI Gym, Pytorch), perception and sensor (LIDAR, MOCAP), and motion control (ROS, Unity).

Relevant Courses

CMU 10703 – Deep Reinforcement Learning and Control
CMU 10708 – Probabilistic Graphical Models
CMU 10716 – Statistical / Advanced Machine Learning
CMU 16720 – Computer Vision
CMU 36700 – Probability and Mathematical Statistics

[1] * indicates equal contribution